Analysis of Attention-span on Prolonged Short-form Online Content Consumption Using Neural Network Based on EEG Signals

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***Abstract*—Electro Encephalogram (EEG) is used to read electrical signals from all portions of the brain by attaching electrodes on the scalp (this research uses 19 channels). These signals are recorded and analyzed on patient that consume large amounts of short form video content and a control patient. We compare brain activity of both patients on their resting states, emotional states, video viewing state and resting state after video viewing. This study studies the trends if any in their brain activity.**

Keywords—Electro Encephalogram (EEG), Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Multi-Layer Perceptron (MLP), Ellipord Filter, Mel-frequency cepstral coefficients (MFCC)

I. Introduction

This research is conducted to analyze the impact of short form content consumption on attention-span. For this study "short attention-span" is defined by the lack of concentration of current task and is seen by the subject's inability to complete tasks and prevalent amount of procrastination. Specifically, the subject of interest in this study are those that meet the criteria above as well as having at least 4 hours of social media usage per day. The subjects were chosen based on a surface level survey by which the participants answered

1. Do they have trouble completing tasks?
2. Do they think they procrastinate a lot?
3. Do they use social medias with short form content at least 4 hours per day.

If a subject answered "yes" for all 3 questions they are considered "experimental patient" as they fit the criteria for which this research is trying to explore. On the other hand, if a participant answered "yes" for all the questions below.

1. Do they use social medias with short form content at least 4 hours per day
2. Do they think they are addicted to social medias with short form content?
3. Do they read (any material) at least 1 hour per day?

They are considered "control patient" and is used to represent the average person as they are not solely reliant on short form content for entertainment. This patient should give baseline EEG readings by which a contrasting EEG reading by experimental patient should highlight the stark differences between brain activity of those that consume a lot of short form content and those that do not.

Short-form content in this research is defined by any online content (mainly videos) that only takes less than 5 minutes to consume and utilises techniques specifically designed to grab the attention of the viewer every 30 seconds. This definition is inspired by Geri et al (2017), in which the article explored the development of such techniques. The results from this study will vary depending on how short form content is defined.

II. Literature Review

A. *Challenging The Six-Minute Myth of Online Video Lectures: Can Interactivity Expand the Attention Span of Learners?*

Geri et al. (2017) conducted research on how to increase attention span of students by adding more interactive elements to teaching materials. The journal illustrates that passive video viewing is the reason students skip class and delay assignments until the last minute and by adding interactive elements of adaptivity, reactivity and flexibility [3] the attention span of students increases dramatically.

The research illustrates the importance of periodically engaging audiences to maintain their focus, which is the tactic used by short form media content as of writing this journal article. However, the side effects of constantly engaging the audience on their attention span overall is not well studied.

The research only proves that the tactics utilised works and not how it might affect or influence the attention span of people.

B. Myth and Mystery of Shrinking Attention Span

An analysis of multiple studies on attention span conducted by Dr. K. R. Subramanian found that attention spans are diminishing due to technological advancements. As marketers, developers and content creators exploit the interactivity of online content, human attention span will continue to decrease. Future generations will most likely adapt to this environment and shorter attention-spans will become the norm.

The study also highlights that this might not be purposeful but instead a side affect of organizations and companies adapting to a changing landscape which in turn further decreases attention span in a vicious cycle. Not much research has specifically focused on the effects of using these attention-grabbing techniques on the brain which is the purpose of this short preliminary study.

III. Objectives

This research specifically aims to measure brain activity and analyse trends (if any) using EEG signals of 19 channels. We will use the overall brain activity to ascertain the patient's

1. Band wave of patient's pre-frontal cortex (level of brain activity) during each activity performed
2. Using overall brain activity to determine emotion of patient during each activity performed

The first objective is to ascertain the level of brain activity the patient experiences during each test that is performed (each test is further elaborated in section 4). The brain activity will act as an indicator on whether the patient is currently focusing or not.

The second objective is to analyse whether emotions play a role in the focus levels of patients. The design of content might trigger certain emotions in patient which might be the cause of the level of brain activity the patient is experiencing.

The purpose of these 2 objectives is to see if there are distinguishing factors between patients who consume a lot of short-form content and those that do not which will act as control patient which theoretically should represent the average person.

IV. Research Protocol

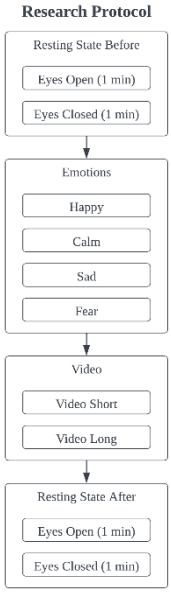

Fig. 1: Research Protocol Flowchart

The research protocol above is used to gather data on the patient's overall brain activity, emotion data which will be used to train a deep learning model to predict the emotion of the patient for the activities performed and their default brain activity.

The resting state is to gauge the level of brain activity of each patient, which will be used as a reference for if there is a significant change in brain activity from the norm.

The emotions are 4 videos that is shown to the subjects and for each emotion their raw EEG data is recorded and later processed to be fed into a neural network to make predictions which should be able to answer objective 2.

The video portion of the protocol consist of 2 videos. 1) short video which is a 1-minute video which utilizes the concepts of grabbing attention [3], 2) a mathematics video which should theoretically be disliked by the majority of the population. The purpose of the second video is to gauge whether the attention-grabbing techniques are the cause of attention spans decreasing. If so when the video is played brain activity of patients should dramatically increase and later decrease when video 2 is shown. If false the brain activity of viewing both videos should relatively be similar.

Lastly the resting state of the participants are measured to see the after effects of having stimulus both positive and negative on both patients.

V. Raw Signal Analysis

The pre-frontal cortex is responsible for decision making, regulating thoughts, emotions and most importantly focus and attention. Using the Ellipord filter to remove white noise and electrical artifacts a clean signal is obtained of which the frequency of channels FP1 and FP2 is analysed. FP1 and FP2 represents the EEG signal from the pre-frontal cortex.

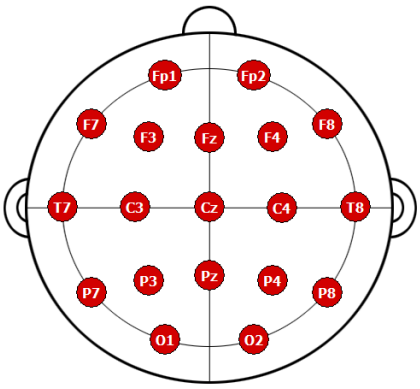

Fig. 2: EEG Channels Used

Using the table below as reference, the frequency of each activity for each of the patient is calculated and recorded. The formula used is as follows.

$$frequency = cycles/time$$

After calculating the frequency of each activity, we can see which level of activity the brain experienced as well as its associated state of mind by following the table below.

TABLE I. BANDWAVE AND ASSOCIATED STATE

Frequency Band Name	Frequency Bandwidth	Associated State
Raw EEG	0 - 45 Hz	Default
Delta	0.5 – 3.5 Hz	Deep Sleep
Theta	4 – 7.5 Hz	Drowsy
Alpha	8 – 12 Hz	Relaxed
Beta	13 – 35 Hz	Engaged

Using the table above as reference, we can map the trend of each patient's brain activity throughout the process of conducting the research protocol. As illustrated in the tables below.

TABLE II. CONTROL PATIENT

Control Patient						
	Eyes Open	Eyes Closed	Video - Short	Video - Long	Eyes Open After	Eyes Closed After
Frequency	10Hz	12.5Hz	10Hz	5hz	10Hz	10Hz
Time	3.4-3.6	3.8-4	0.4-0.6	4.4-4.6	2.8-3	0.2-0.4
Band	Alpha	Beta	Alpha	Theta	Alpha	Alpha

TABLE III. EXPERIMENTAL PATIENT

Experimental Patient						
	Eyes Open	Eyes Closed	Video - Short	Video - Long	Eyes Open After	Eyes Closed After
Frequency	5Hz	7.5Hz	12.5Hz	10Hz	6.7Hz	8.3Hz
Time	0.7-0.9	3.4-3.8	1.4-1.6	3.8-4	3.4-3.7	1.7-2
Band	Theta	Theta	Beta	Alpha	Theta	Alpha

From the tables above, a graph can be produced to clearly show the level of brain activity for both patients when undergoing the protocol.

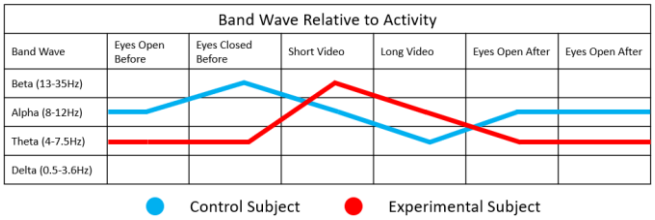

Fig. 3: Level of Brain Activity

By organising the order of band wave from highest to lowest and graphing the activity, a trend can be shown on the state of the 2 patient's level of brain activity. The lower the line on the graph the "sleepier" or slower the brain activity as Delta band waves is associated with deep sleep while Beta is associated with focus and engagement.

The graph shows that the control subject's (blue line) default state is within the band wave of alpha and beta which according to table 1 means that the control subject is in a state of either relaxation or focus. After showing the short video the control subject's band wave stays within the alpha range meaning the attention-grabbing content does not seem to significantly affect his band wave level i.e., it is no different than when the subject is in resting state. On the other hand, when the long video was shown, a dip in the band wave occurs, dropping to Theta which indicates that the subject is beginning to become sleepy, this is probably due to the content of the video containing mathematics which for a majority of people is a subject matter of low interest. Furthermore, the subjects band wave returns to his default resting state maintaining within the alpha range. The main takeaway from the control subjects would be that the control subject only loses focus or engagement when a stimulus of great disinterest is shown and their default state is relaxed and is actively focused.

The red line illustrates the default state of mind for the experimental subject, compared to control subject, the experimental subject's resting state falls withing the theta range, which means the overall brain activity indicates that the subject is either drowsy or is merely not focused or relaxed compared control subject. This might suggest that the experimental subject isn't consciously engaged with their actions. As of the time of recording the data, both subjects have not had any sleep scheduling issues and is well rested. Another activity which supports the previously stated sentiment is the spike in band wave category to beta when shown the short video which includes the attention-grabbing techniques. It slightly drops in band wave when the long video was shown however a caveat to this particular piece of information is that the experimental subject is fairly fond of mathematics thus his level of brain activity did not fall as we had theorised it would to default state of theta. After both stimulus (videos) was shown the experimental subject immediately returned to default state of Theta. This could indicate that the subject might have been desensitized to stimulus in general and the only thing that piques their interest might be hyperstimulation which the videos do provide and can be seen in the sudden spike in band wave all the way to Beta. Whether this is due to desensitization or another factor has to be researched further, this research is to merely show

that there might be a grounds for exploring such research by showing whether there is a trend or not.

A sample of both subject's filtered EEG signals is provided down below. The only readings that were considered were channels FP1 and FP2 for readings of pre-frontal cortex.

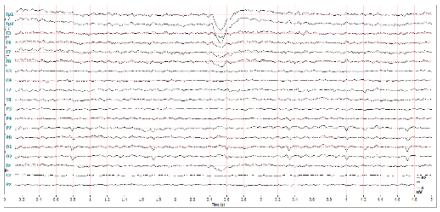

Fig. 4.1: Control Subject Eyes Open

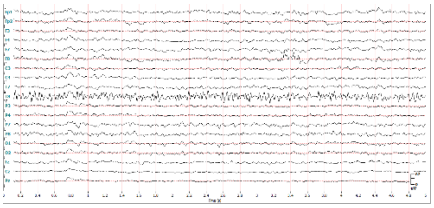

Fig. 4.2: Experimental Subject Eyes Open

VI. Analysis of Emotions

The emotional data of both patients are recorded by showing them videos that are supposed to be "happy", "calming", "sad" and "scary" which is supposed to invoke emotions of happiness, calm, sad and fear within the patients.

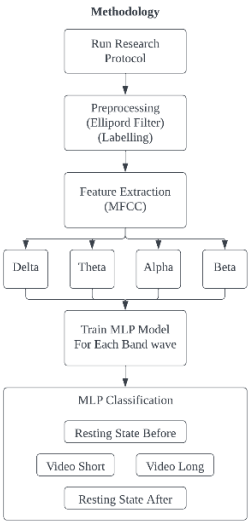

Fig. 4.3: Methodology for Predicting Emotion

The emotional data highlighted in the research protocol is compiled and using MFCC feature extraction technique we can extract specific band waves of Delta, Theta, Alpha and Beta for each activity the subjects undergo. Each band wave of each activity is fed into a Multi-layer Perceptron, its structure is illustrated below.

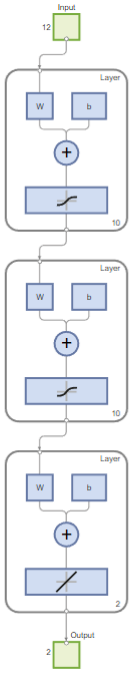

Fig. 5: Multi-layer Perceptron Topography

The MLP used for this experiment consists of:

1. 12 Nodes Input Layer
2. 2x10 Nodes Hidden Layer (Tanh Activation Function)
3. 1x10 Nodes Hidden Layer (Linear Activation Function)
4. 2 Nodes Output Layer (Linear Activation Function)

The emotional data is labelled based on Russel's model of using valence and arousal, the data is labelled as follows.

TABLE IV. Russel's Model

Emotion	Valence	Arousal
Happy	1	1
Calm	1	-1
Sad	-1	-1
Fear	-1	1

Each band wave for each activity is fed into the same MLP model and using the results of the prediction a compilation of the results of each patient's emotion for resting state, video and resting state after video viewing is shown below.

The emotion of each activity may vary within the test itself and thus must be manually reviewed by graphing the emotion of each prediction graph. An example of the variability of emotions within the same action is illustrated in the figure below.

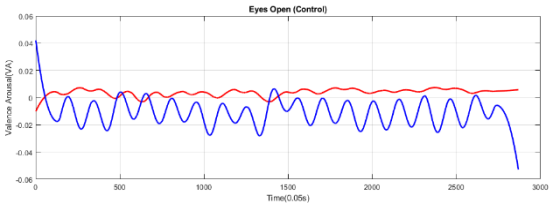

Fig. 6: Example of Emotion Variability for One Activity

The graph above is an example of analysing emotion by using Russel's model. The red line represents valence and the blue represents arousal. When the line is below the y axis it is considered -1 and vice versa. The emotion is determined by the majority of the emotion for each sector in increments of 500 (total for each graph is 3000 meaning 6 sectors).

After compiling the dominant emotion for each band wave for each activity within the protocol, a table of emotions is produced as seen down below.

TABLE V. EMOTIONS OF CONTROL PATIENT

Control Patient						
	Eyes Open Before	Eyes Closed Before	Video - Short	Video - Long	Eyes Open After	Eyes Closed After
Alpha	Calm	Calm	Sad	Calm	Calm	Sad/Calm
Beta	Happy	Happy	Happy	Fear	Fear/Happy	Happy
Delta	Sad	Fear/Sad	Happy/Fear	Sad	Sad	Calm
Theta	Happy	Calm	Sad	Fear	Happy	Fear
Results	Happy	Calm	Happy	Fear	Happy	Calm

TABLE VI. EMOTIONS OF EXPERIMENTAL PATIENT

Experimental Patient						
	Eyes Open Before	Eyes Closed Before	Video - Short	Video - Long	Eyes Open After	Eyes Closed After
Alpha	Calm	Calm	Sad	Calm	Sad	Sad/Calm
Beta	Happy/Fear	Calm	Happy	Happy	Happy	Calm
Delta	Happy	Sad	Happy	Happy	Happy/Sad	Sad
Theta	Happy	Fear	Happy	Happy	Sad	Happy
Results	Happy	Calm	Happy	Happy	Sad	Calm/Sad

The last rows of the two emotion tables above illustrates that the emotions between control patient and experimental patient are negligible and is not significant enough to draw a definite conclusion on whether emotional state plays a role in affecting attention length. If tables V (5) and VI (6) are cross referenced with figure 3, it is clear that the emotions of both patients are similar even though the band wave level of both patients are at different as seen by alpha waves by control patient and theta waves by experimental patient.

The only notable difference is that control patient is feeling fear while viewing video with mathematics and experimental is feeling happy probably due to their fondness over such material, however, this is not enough evidence to state that emotion affect the amount of concentration. Another notable difference in the emotion is the eyes open after viewing long video. The control patient is happy after a disliked video has ended while the experimental patient exhibited sadness, which might indicate their disappointment when stimulus is taken away. However, these are not concrete enough evidence as to determine whether it affects attention span which is represented by either a relaxed or engaged/focus band wave, of which the correlation does not exist and a larger sample size must be obtained.

VII. CONCLUSION

In conclusion section 5 shows there is a trend present in which the experimental subject showed an overall slower brain activity exhibited by their slower band wave reading of the pre-frontal cortex and it only increased when a stimulus is introduced, the increase was also very significant as seen in figure 3 where the patient's initial resting state went from Theta which represents drowsiness and jumping 2 band waves to Beta which represents focus. On the other hand, the control patient did not exhibit such drastic changes in brain activity and only decreased when shown stimulus in which they disliked, however, the control patient's overall brain activity was much healthier as humans should exhibit a relaxed state when no stimulus is applied.

The emotional results did not show any correlation between emotional state and levels of brain activity as the emotions the subjects are feeling are similar even though their default and reacted levels of brain activity is completely different which indicates that emotion does not play a large role is determining one's attention span. A larger sample size is needed to verify this result.

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